FIITJEE

ALL INDIA INTEGRATED TEST SERIES

PART TEST – III

JEE (Advanced)-2021

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

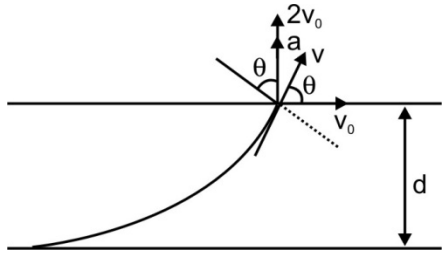
PART – I

SECTION – A

1. D

Sol. $a\cos\theta = \frac{v^2}{R}$

(Where $v = \sqrt{(2v_0)^2 + v_0^{2-}}$ $a = \frac{qE}{m}$ $\tan\theta = \frac{2v_0}{v_0}$)



2. A

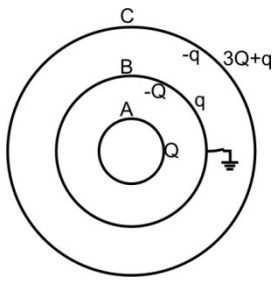
Sol. $v_B = 0$

$v_B = \frac{KQ}{2R} + \frac{K(q-Q)}{2R} + \frac{K3Q}{3R}$

$\frac{Q}{2} + \frac{(q-Q)}{2} + Q = 0$

$2Q + q = 0$

$q = -2Q$



3. B

Sol. From energy conservation

From P + 0 Q

$V = \sqrt{2gR\sin\theta}$

Force, equation into radial direction

$N + qvB - mg\sin\theta = \frac{mv^2}{R}$

$N = \left[3mg\sin\theta - qB\sqrt{2gR}\right]$

4. A

Sol. Potential at surface should be zero

$$\frac{KQ}{2a}+\frac{kq}{a}=0$$

$$q=\frac{-Q}{2}$$

5. B

Sol. $$\frac{x}{R}=\frac{52-2}{48-1}$$

$X = 10.6\ \Omega$

6. C

Sol. Total charge inside the square $\frac{2q}{3}$

$$\text{Flux} = \frac{q_{in}}{\in} = \frac{2q}{3\in_0}$$

7. B

Sol. Balance length is independent of radius of the wire.

8. ABC

Sol. At steady state, capacitor will offer infinite resistance.

9. ABC

Sol. $$T=\frac{2u\sin\theta}{g},\ H=\frac{u^2\sin^2\theta}{2g}$$

$$R=u\cos\theta T+\frac{1}{2}\frac{qE}{m}T^2$$

10. AC

Sol. (A) For an electron, $\frac{\text{Magnetic momentum}}{\text{Angular momentum}}=\frac{e}{2m_e}$

$$\frac{M}{L}=\frac{e}{2m_e}$$

According to Bohr's theory, $L=n\frac{h}{2\pi}=\frac{h}{2\pi}$ (for ground state)

$$\therefore \quad M=\left(\frac{e}{2m_e}\right)L=\frac{e}{2m_e}\frac{h}{2\pi}=\frac{eh}{4\pi m_e}$$

(C) Torque $=MB\sin30^\circ=\frac{ehB}{8\pi m_e}$

11. AD

Sol. *Use;* $B=\frac{\mu_0}{2}\left(\vec{J}\times\vec{r}\right)$

12. ABD

Sol. $\omega_{Au} = \Delta K$

$\omega_E + \omega_B = \Delta K$ *but* $\omega_B = 0$

$E \cdot 2a = \frac{1}{2} m(2v)^2 - \frac{1}{2} mv^2$

Rate of work done at $P = f \cdot v = qEv$

Rate of work done at $Q = 0$ *as* $\vec{f} \perp \vec{v}$

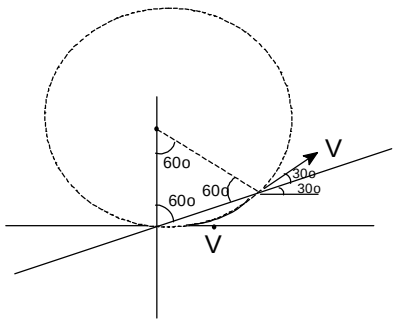
13. BC

Sol. $\vec{v} = v\cos 60\,\hat{i} + v\sin 60\,\hat{j}$

$\omega = \frac{qB}{m}$

$t = \frac{\pi/3}{\omega}$

$Dis\tan ce = R \times \theta$



14. CD

Sol. $B(2\pi r) = \mu_o \int_o^r J_o \left(\frac{r}{R}\right) 2\pi r dr$

15. C

Sol. $\frac{q_1}{2c} = \frac{q_2}{c}$

& $q_1 + q_2 = 5cv$

$q_1 = \frac{10cv}{3}$ & $q_2 = \frac{5cv}{3}$

$P.d.\ across\ each = \frac{5v}{3}$

$equivalent\ time\ count = R \cdot \frac{2c \times c}{2c + c} = \frac{2Rc}{3}$

16. D

Sol. $\frac{q_1}{2c} = \frac{q_2}{c}$

& $q_1 + q_2 = 5cv$

$q_1 = \frac{10cv}{3}$ & $q_2 = \frac{5cv}{3}$

$P.d.\ across\ each = \frac{5v}{3}$

$equivalent\ time\ count = R \cdot \frac{2c \times c}{2c + c} = \frac{2Rc}{3}$

17. D

Sol. $i_1 = \dfrac{E_1}{R_1 + R_{AB}} = \dfrac{4}{25}$

$V_{AB} = i_1 \times R_{AB} = \dfrac{40}{25} = \dfrac{8}{5} = 1.6$

$k = \dfrac{V_{AB}}{\ell} = \dfrac{1.6}{50}\ V/cm$

$E_2 = \dfrac{1.6}{50} \times 31.25 = 1V$

Again, when both S. are closed

$K_2 = \dfrac{4}{10} \times \dfrac{10}{50} = \dfrac{4}{50}$

$\dfrac{E_2}{5+r} \times 5 = \dfrac{4}{50} \times 5$

$20 + 4r = 50$

$r = \dfrac{30}{4} = 7.5 \quad (D)$

Again when K_2 open K_1 is closed for balance length.

$E_2 = K_2 \cdot \ell \Rightarrow \ell = \dfrac{E_2}{K_2} = \dfrac{50}{4} \quad (C)$

18. C

Sol. $i_1 = \dfrac{E_1}{R_1 + R_{AB}} = \dfrac{4}{25}$

$V_{AB} = i_1 \times R_{AB} = \dfrac{40}{25} = \dfrac{8}{5} = 1.6$

$k = \dfrac{V_{AB}}{\ell} = \dfrac{1.6}{50}\ V/cm$

$E_2 = \dfrac{1.6}{50} \times 31.25 = 1V$

Again, when both S. are closed

$K_2 = \dfrac{4}{10} \times \dfrac{10}{50} = \dfrac{4}{50}, \quad \dfrac{E_2}{5+r} \times 5 = \dfrac{4}{50} \times 5$

$20 + 4r = 50$

$r = \dfrac{30}{4} = 7.5 \quad (D)$

Again when K_2 open K_1 is closed for balance length.

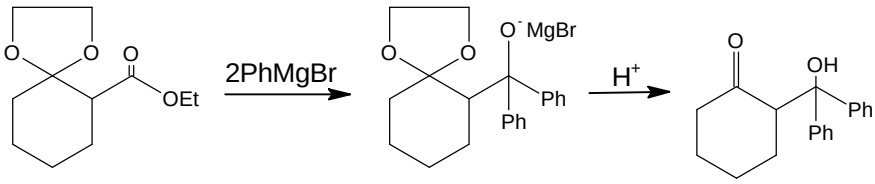
$E_2 = K_2 \cdot \ell \Rightarrow \ell = \dfrac{E_2}{K_2} = \dfrac{50}{4} \quad (C)$

Chemistry

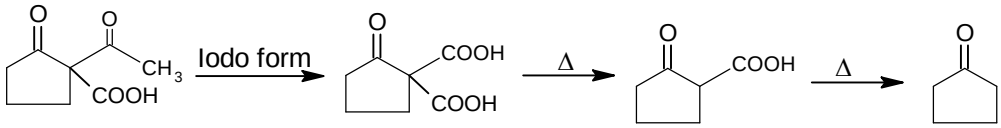
PART – II

SECTION – A

19. A
Sol. A =



20. C
Sol.



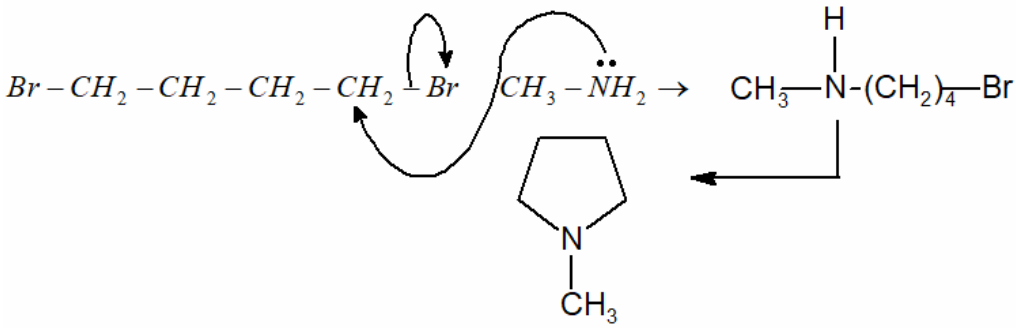
21. A
Sol. $NaHCO_3$ $Ph-COOH$ will give CO_2 *with* $NaHCO_3$ while $Ph-CH_3$ not.

22. B
Sol. HIO_4 will not deave ether.

23. C
Sol. Conceptual

24. D
Sol. Acetaldehyde give positive test with NaOI and fehling reagent while Ph – CHO do not.

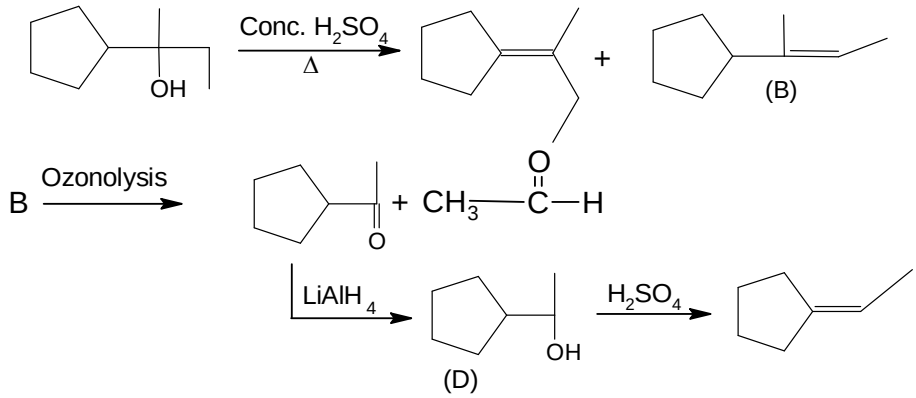
25. B
Sol.



26. BC
Sol. Conceptual

27. BD

Sol.

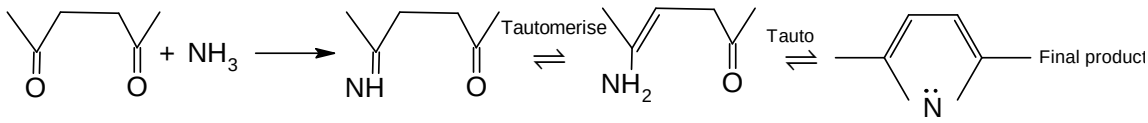


28. BCD

Sol. Conceptual

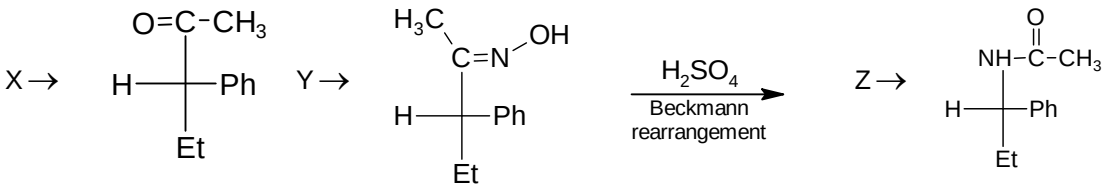
29. AB

Sol.



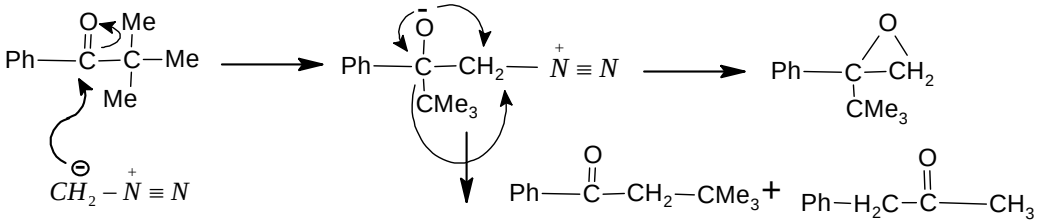
30. ABC

Sol.



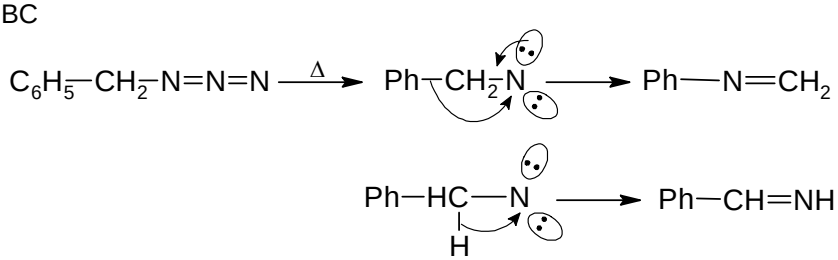
31. ABC

Sol.

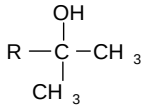


32. BC

Sol.

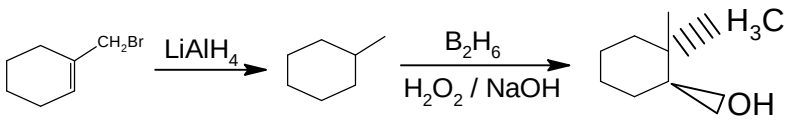


33. D

Sol. B is an alcohol (3) which have  type skelton

R is C_7H_{11} WITH d.u = 2 so it must be Cyclohexenyl methyl

34. B

Sol. 

35. B

Sol. $Na_2\left[Fe(CN)_5 \cdot NO\right] + Na_2S \rightarrow Na_4\left[Fe(CN)_5 \cdot NOS\right]$

36. A

Sol. Lassaigne extract contains $CN^{\ominus}$, $CNS^{\ominus}$, S^{2-}

Mathematics

PART – III

SECTION – A

37. A

Sol. $f(g(x)) = x$

$[f'(g(x))].g'(x) = 1$

$g'(x) = \frac{1}{f'(g(x))}$

$g'\left(-\frac{7}{6}\right) = \frac{1}{f'(1)}$

$f(1) = -\frac{7}{6}$

$\Rightarrow g\left(-\frac{7}{6}\right) = 1$

38. B

Sol. $\sin^3 x = \frac{3}{4}\sin x - \frac{1}{4}\sin 3x$

$\therefore \int_0^\infty \frac{\sin^3 x}{x}dx = \frac{3}{4}\int_0^\infty \frac{\sin x}{x}dx - \frac{1}{4}\int_0^\infty \frac{\sin 3x}{x}dx$

$= \frac{3}{4}\int_0^\infty \frac{\sin x}{x}dx - \frac{1}{4}\int_0^\infty \frac{\sin u}{u}du \qquad (u = 3x)$

$= \frac{3}{4}\frac{\pi}{2} - \frac{1}{4}\frac{\pi}{2} = \frac{\pi}{4}$

39. D

Sol. $\sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cos\left(x + \frac{\pi}{3}\right)$

$= \sin^2 x + \left(\frac{\sin x}{2} + \frac{\sqrt{3}\cos x}{2}\right)^2 + \cos x\left(\frac{\cos x}{2} - \frac{\sqrt{3}\sin x}{2}\right)$

$= \sin^2 x + \frac{\sin^2 x}{4} + \frac{3\cos^2}{4} + \frac{\cos^2 x}{2}$

$= \frac{5\sin^2 x}{4} + \frac{5\cos^2 x}{4}$

$= \frac{5}{4}$

40. A

Sol. F(x) is continuous at some x where sin x = cos x or tan x = 1 or $x = n\pi + \frac{\pi}{4}$, $n \in I$.

41. C

Sol. For $f(x)$ to be continuous at $x = 0$, we have

$f(0^-) = f(0^+)$ or $a(0) + b = 1$ or $b = 1$

$f'(0^+) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{e^{h^2+h} - b}{h}$

$= \lim_{h\to \infty} \frac{e^{h^2+h} - 1}{h} = \lim_{h\to 0} \frac{e^{h^2+h} - 1}{h(h+1)}(h+1) = 1$

$\therefore f'(0^-) = a$

Hene, $a = 1$.

42. B

Sol. $x^2 - y^2 = 8$

$\therefore \frac{dy}{dx} = \frac{x}{y}$ or $\frac{1}{dy/dx} = \frac{y}{x}$

At the point $\left(-\frac{5}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$, $-\frac{1}{dy/dx} = \frac{-\frac{3}{\sqrt{2}}}{-\frac{5}{\sqrt{2}}} = \frac{3}{5}$

Also, $9x^2 + 25y^2 = 225$

$\therefore 18x + 50y\frac{dy}{dx} = 0$

or $\frac{dy}{dx} = -\frac{9x}{25y}$ or $-\frac{dx}{dy} = \frac{25y}{9x}$

At the point $\left(-\frac{5}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$,

$-\frac{dx}{dy} = \frac{25 \times \frac{3}{\sqrt{2}}}{9\left(-\frac{5}{\sqrt{2}}\right)} = -\frac{15}{9} = -\frac{5}{3}$

43. A

Sol. Since the same line is tangent at one point x = a and normal at other pint x = b, tangent at x = b will be perpendicular to tangent at x = a.

Also, slope of tangent changes from positive to negative or negative to positive. Therefore, it takes the value zero somewhere. Thus, there exists a point $c \in (a, b)$ where $f'(c) = 0$.

44. ABC

Sol. $f(x) = \frac{\sin(x+a)}{\sin(x+b)}$

$f'(x) = \frac{\sin(x+b)\cos(x+a) - \sin(x+a)\cos(x+b)}{\sin^2(x+b)}$

$= \frac{\sin(b-a)}{\sin^2(x+b)}$

If $\sin(b-a) = 0$, then $f'(x) = 0$ or $f(x)$ will be constant,

i.e., $b - a = n\pi$ or $b - a = (2n+1)\pi$ or $b - a = 2n\pi$.

Then $f(x)$ has no minima.

45. AC

Sol. $xy = (a+x)^2$

or $y + xy' = 2(a+x)$

Now, $y' = \pm 1$

or $y \pm x = 2(a+x)$

$\frac{(a+x)^2}{x} \pm = 2(a+x)$

or $\pm x = 2(a+x) - \frac{(a+x)^2}{x}$

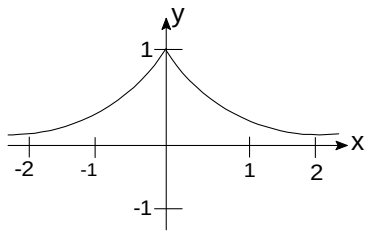
or $\pm x^2 = (a+x)(x-a)$

or $\pm x^2 = x^2 - a^2$

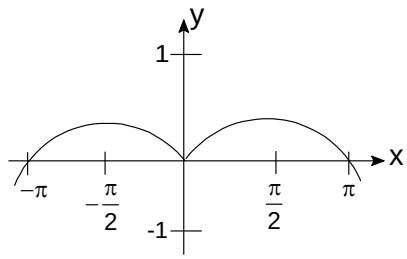
or $2x^2 = a^2$ or $x = \pm \frac{a}{\sqrt{2}}$

46. ABD

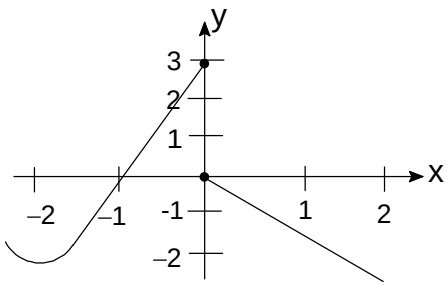
Sol. Graph of $y = e^{-|x|}$



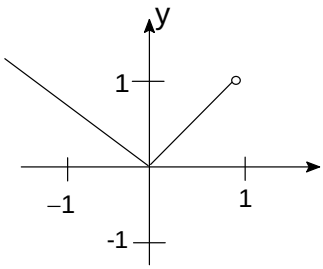
Graph of $y = \sin|x|$



$f(x) = \begin{cases} x^2 + 4x + 3, & x < 0 \\ -x, & x \geq 0 \end{cases}$



$$f(x)=\begin{cases}|x|, & x<0\\ \{x\}, & x\geq 0\end{cases}$$



47. ACD

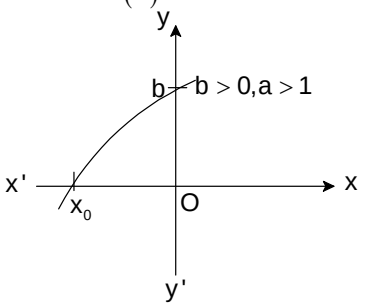
Sol. (A) is not correct as f(x) = x from R to R is onto but its reciprocal function $g(x)=\frac{1}{f(x)}$ is not onto on R.

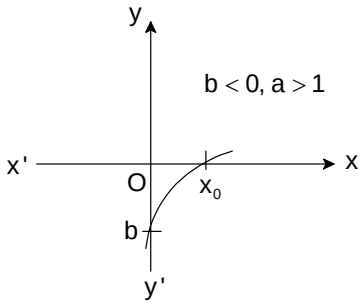
Option (B) is obviously true.

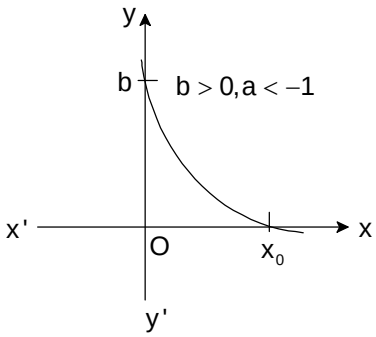
Also, g(x) is not continuous and, hence, not differentiable through f(x) is continuous and differentiable in the above case.

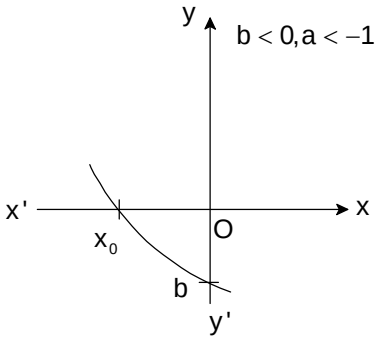
48. ABC

Sol. $f'(x)=\cos x+a$









If a > 1, then $f'(x)=0$ or f(x) is an increasing function. Then f(x) = 0 has +ve root if b < 0 and –ve root if b > 0.

If a < -1, then $f'(x)<0$ or f(x) is a decreasing function. Then f(x) = 0 has negative root if b < 0.

49. BC

Sol. Since g(x) is increasing and f(x) is decreasing.

$g(x+1)>g(x-1)$ and $f(x+1)<f(x-1)$

or $f\{g(x+1)\}<f\{g(x-1)\}$

and $g\{f(x+1)\}<g\{f(x-1)\}$

50. AD

Sol. $f(x+\pi)=\int_0^{x+\pi}\left(\cos(\sin t)+\cos(\cos t)\right)dt$

$=\int_0^{\pi}\left(\cos(\sin t)+\cos(\cos t)\right)dt$

$+\int_{\pi}^{x+\pi}\left(\cos(\sin t)+\cos(\cos t)\right)dt$

$=f(\pi)+\int_0^{x}\left(\cos(\sin t)+\cos(\cos t)\right)dt$

$\left[\because \text{for } g(x)=\cos(\sin x)+\cos(\cos x), f(x+\pi)=f(x)\right]$

$=f(\pi)+f(x)$

$=f(\pi)+2f\left(\frac{\pi}{2}\right)$ $\qquad\qquad \left[\because g(x) \text{ has period } \frac{\pi}{2}\right]$

51. D

Sol. $\int_2^{x} f(t)dt=\frac{x^2}{2}+\int_x^{2} t^2 f(t)dt$

Differentiating w.r.t. x, we get

$f(x)=x+\left(-x^2 f(x)\right)$

or $f(x)=\frac{x}{1+x^2}$

or $yx^2-x+y=0$

Since x is real, $D\geq 0$

or $1-4y^2\geq 0$

or $y\in\left[-\frac{1}{2},\frac{1}{2}\right]$

Also, $f(x)$ is an odd function. Hence, $\int_{-2}^{2} f(x)dx=0$

$f'(x)=\frac{1+x^2-2x^2}{1+x^2}=\frac{1-x^2}{1+x^2}\geq 0$

or $x^2-1\leq 0$

or $x\in[-1,1]$

52. A

Sol. $\int_2^{x} f(t)dt=\frac{x^2}{2}+\int_x^{2} t^2 f(t)dt$

Differentiating w.r.t. x, we get

$f(x)=x+\left(-x^2 f(x)\right)$

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Also, $f(x)$ is an odd function. Hence, $\int_{-2}^{2} f(x)dx = 0$

$f'(x) = \frac{1+x^2-2x^2}{1+x^2} = \frac{1-x^2}{1+x^2} \geq 0$

or $x^2-1 \leq 0$ or $x \in [-1, 1]$

53. C

Sol. $L = \lim_{x\to\infty}\left(x+1-\sqrt{ax^2+x+3}\right)$

$= \lim_{x\to\infty}\left(\frac{(x+1)^2-\left(ax^2+x+3\right)}{x+1+\sqrt{ax^2+x+3}}\right)$

$= \lim_{x\to\infty}\left(\frac{(1-a)x^2+x-2}{x+1+\sqrt{ax^2+x+3}}\right)$

L exists finitely if $1-a=0$ or $a=1$

$\therefore L = \lim_{x\to\infty}\left(\frac{x-2}{x+1+\sqrt{x^2+x+3}}\right)$

$= \lim_{x\to\infty}\left(\frac{1-\frac{2}{x}}{1+\frac{1}{x}+\sqrt{1+\frac{1}{x}+\frac{3}{x^2}}}\right)$

$= \frac{1}{2}$

54. B

Sol. $L = \lim_{x\to\infty}\left(x+1-\sqrt{ax^2+x+3}\right)$

$= \lim_{x\to\infty}\left(\frac{(x+1)^2-\left(ax^2+x+3\right)}{x+1+\sqrt{ax^2+x+3}}\right)$

$= \lim_{x\to\infty}\left(\frac{(1-a)x^2+x-2}{x+1+\sqrt{ax^2+x+3}}\right)$

L exists finitely if $1-a=0$ or $a=1$

$\therefore L = \lim_{x\to\infty}\left(\frac{x-2}{x+1+\sqrt{x^2+x+3}}\right)$

$= \lim_{x\to\infty}\left(\frac{1-\frac{2}{x}}{1+\frac{1}{x}+\sqrt{1+\frac{1}{x}+\frac{3}{x^2}}}\right)$

$= \frac{1}{2}$